NIGHITHA T N

DATA SCIENCE & AI PROFESSIONAL

Kerala, India | +91 6282937620 | nighithat@gmail.com | LinkedIn | GitHub

---

PROFESSIONAL SUMMARY

Data Science and AI professional with expertise in Machine Learning, Statistical Analysis, and Python-based development. Skilled in building and optimizing predictive models using scikit-learn and TensorFlow, with strong experience in exploratory data analysis and feature engineering. Adept at handling structured and unstructured datasets to design scalable, data-driven solutions.

TECHNICAL SKILLS

Programming & Databases: Python | R | SQL | Linux
Machine Learning & Modeling: Scikit-learn | Supervised Learning | Unsupervised Learning | Linear Regression | Logistic Regression |Decision Trees |Random Forest | Classification | Regression | Model Evaluation | CrossValidation | Hyperparameter Tuning
Deep Learning & NLP: TensorFlow | PyTorch (Basic)| Keras | Neural Networks | ANN | CNN | RNN | Text Preprocessing | TF-IDF | Sentiment Analysis
LLMs & Generative AI: Hugging Face Transformers | LangChain | LLM Fine-tuning | Prompt Engineering | RAG | Text Embeddings
Data Analysis & Feature Engineering: Pandas | NumPy | Data Cleaning | Data Preprocessing | Feature Engineering | EDA | Statistical Analysis
Model Deployment & APIs: Streamlit | FastAPI | Flask | REST API Development | Model Integration
Data Visualization & Reporting: Matplotlib | Seaborn | Power BI | Tableau | Advanced Excel
Tools & Version Control: Jupyter Notebook | Anaconda | Git | GitHub | Basic MLOps Practices

---

PROFESSIONAL EXPERIENCE

Data Science Intern | Spinnaker Analytics, Boston, USA Feb 2026 - Present

- Contributing to data science and AI research projects involving data analysis and machine learning workflows
- Performing data preprocessing, exploratory data analysis (EDA), and feature engineering using Python
- Assisting in model development and evaluation for predictive analytics tasks
- Supporting research experiments and documenting analytical insights

Project Intern | Eminent Biosciences Pvt Ltd., Indore, India Feb 2025 - Aug 2025

- Completed M.Sc. dissertation on transcriptomic profiling using large-scale RNA-seq datasets
- Performed differential gene expression analysis, statistical testing, and biological interpretation
- Contributed to multiple research projects involving NGS data processing and analysis
- Co-authored AI-focused book chapters on genomics, drug repurposing, and medical systems
- Participated in team discussions and followed structured development workflows

---

PROJECTS

AI Resume Screening & Talent Match System (Ongoing)

- Developing an NLP-based resume screening and job-candidate matching system
- Implementing text preprocessing, embeddings, and similarity scoring
- Designing candidate ranking logic based on skill alignment
- Building an Android application integrated with a backend ML API

Fraud Detection in Financial Transactions (ML + Power BI + Flask) | GitHub

- A binary classification model was built to detect fraudulent financial transactions
- Carried out data preprocessing, feature scaling, and class imbalance handling
- Evaluated model performance using precision, recall, F1-score, and confusion matrix
- Improved an interactive dashboard for real-time fraud risk insights

Tools used: Python, Scikit-learn, Pandas, NumPy, Power BI, Flask, Matplotlib

Customer Churn Prediction (ML + Power BI + Streamlit) | GitHub

- Built classification models using Logistic Regression and Random Forest
- Performed data preprocessing, feature engineering, and class imbalance handling
- Optimized models using cross-validation and hyperparameter tuning
- Deployed an interactive prediction dashboard using Streamlit

Tools used: Python, Scikit-learn, Pandas, NumPy, Power BI, Streamlit, Matplotlib

Twitter Sentiment Analysis (NLP + FastAPI + Power BI + BiLSTM) | GitHub

- Developed an NLP pipeline with text preprocessing and TF-IDF vectorization
- Implemented a BiLSTM model for sentiment classification
- Built a REST API using FastAPI for real-time inference
- Evaluated model performance using accuracy and a confusion matrix

Tools used: Python, TensorFlow, Keras, FastAPI, Pandas, NumPy, Power BI

Smart Retail Analytics (ML + Streamlit Deployment) | GitHub

- A machine learning model for sales prediction and demand forecasting was built
- Executed data preprocessing, feature engineering, and model evaluation
- Applied regression techniques to optimize prediction performance
- Enhanced an interactive Streamlit dashboard for real-time business insights

Tools used: Python, Scikit-learn, Pandas, NumPy, Streamlit, Matplotlib

---

EDUCATION

- **Diploma in Data Science & AI** — Oct 2025 - Feb 2026
  Boston Institute of Analytics
- **Master of Science in Bioinformatics (Specialization in NGS Data Analytics)** — 2023 - 2025
  University of Kerala
- **Bachelor of Science in Biochemistry** — 2018 - 2021
  University of Kerala

PUBLICATIONS

- Co-author, AI for Emotion and Sentiment Analysis. **ISBN: 9789391700645**
- Co-author, AI-Powered Genomic Medicine. Springer Nature, 2025. **ISBN: 978-0-443-44415-9**
- Co-author, AI-Driven Drug Repurposing and Combination Therapies. Springer (In Press)
- Co-author, IoMT-based Smart Emergency Medical Systems. Springer (In Press)

CERTIFICATIONS

- **Gen AI Engineering Mastermind** | Outskill — 2026
- **R Programming** | Coursera - Johns Hopkins University, **Credential ID: IXUI4O1LEY2E** — 2025
- **NGS Data Analysis** | Eminent Biosciences — 2025
- **Python Programming** | Eminent Biosciences — 2025
- **Bioinformatics & Drug Designing** | Eminent Biosciences — 2024

WORKSHOPS & TRAINING

- **Agentic AI Workshop** | Techvantage.ai — 2025
- **Advanced AWK Programming for NGS Data Analysis** | Eminent Biosciences — 2025

LANGUAGES

- **English** (Fluent)
- **Hindi** (Fluent)
- **Malayalam** (Fluent)
- **Tamil** (Fluent)
- **French** (Basic)